Conic Particle Methods for Mixed Nash Equilibria

Synthesizing WFR Geometry with Slingshot GDA

All experiments are implemented as Jupyter notebooks available in the GitHub repository. Animations are in the `gifs/` directory of this project as well as the repo. The two notebooks are:

- `CPMDA_Slingshot.ipynb`
- `CPMP_Slingshot.ipynb`

Contents

1 Motivation and Problem Setting

We consider a continuous, two-player, zero-sum game. Player 1 (minimizer) chooses a strategy $x \in \mathcal{X} \subseteq \mathbb{R}^d$ and Player 2 (maximizer) chooses $y \in \mathcal{Y} \subseteq \mathbb{R}^d$. The payoff to Player 2 is $f(x, y)$, so Player 1 wishes to minimize and Player 2 wishes to maximize the same quantity.

In a **mixed strategy** formulation, each player commits to a *probability measure* over their action space rather than a single deterministic action. The equilibrium objective becomes:

$$\min_{\mu \in \mathcal{P}(\mathcal{X})} \max_{\nu \in \mathcal{P}(\mathcal{Y})} F(\mu, \nu) := \mathbb{E}_{(x,y)\sim\mu\otimes\nu}\big[f(x, y)\big].$$

A pair (μ^*, ν^*) satisfying

$$F(\mu^*, \nu) \leq F(\mu^*, \nu^*) \leq F(\mu, \nu^*) \quad \forall\, \mu, \nu$$

is called a **Mixed Nash Equilibrium (MNE)**.

1.1 Why Standard GDA Fails

Gradient Descent-Ascent (GDA) iterates $x \leftarrow x - \alpha\nabla_x f$, $y \leftarrow y + \beta\nabla_y f$ simultaneously. On the canonical bilinear game $f(x, y) = xy$, the continuous-time flow is purely rotational — it orbits the saddle point forever and never converges. Discretization does not help; the iterates typically spiral *outward* and diverge. This rotational instability is the central pathology motivating both works synthesized here.

2 The Conic Particle Framework [Wang and Chizat, 2025]

2.1 Atomic Measure Parameterization

Rather than searching over all probability measures (an infinite-dimensional space), we parameterize each player's strategy as a finite **atomic measure**:

$$\mu = \sum_{i=1}^{n} a_i\, \delta_{x_i}, \qquad \nu = \sum_{j=1}^{m} b_j\, \delta_{y_j},$$

where δ_p is the Dirac mass at p. The constraints $a_i > 0$, $\sum_i a_i = 1$ (and similarly for b_j) ensure these are valid probability measures. Each particle (a_i, x_i) represents a pure strategy x_i played with probability a_i.

The payoff becomes a finite sum:

$$F(\mu, \nu) = \sum_{i=1}^{n}\sum_{j=1}^{m} a_i b_j\, f(x_i, y_j).$$

2.2 The Wasserstein–Fisher–Rao Geometry

The key insight of Chizat is to optimize in the **Wasserstein–Fisher–Rao (WFR)** metric, which is the natural Riemannian geometry on the space of positive measures. It couples two classical geometries:

- **Fisher–Rao (FR) geometry** acts on the *weights* a_i. It is the information geometry of probability simplices, where the natural gradient is the multiplicative update. It governs how probability mass is *reallocated* between particles.

- **Wasserstein (W) geometry** acts on the *positions* x_i. It is the optimal transport geometry on $\mathbb{R}^d$, where the natural gradient corresponds to moving particles through space. It governs how pure strategies are *refined*.

Definition 1 (WFR Gradient)**.** The WFR gradient of F with respect to $\mu = \sum_i a_i \delta_{x_i}$ has two components at each particle i:

$$\underbrace{\frac{\partial F}{\partial a_i}}_{\text{FR component}} = \sum_{j=1}^{m} b_j\, f(x_i, y_j), \tag{1}$$

$$\underbrace{\frac{\partial F}{\partial x_i}}_{\text{W component}} = \sum_{j=1}^{m} b_j\, \nabla_x f(x_i, y_j). \tag{2}$$

The FR component tells each particle how good its current position is on average (weighted by opponent's strategy). The W component tells each particle which direction to move to improve.

Remark 1. Equation (1) is the expected payoff of particle i against the opponent's current mixed strategy ν. Particles with high expected payoff (good positions for P1) will *lose* weight; particles with low expected payoff will *gain* weight — this is mirror descent on the simplex.

2.3 The Mirror Descent Weight Update

Operating in the FR geometry, the natural update for weights is **Mirror Descent** with the negative entropy mirror map $\phi(a) = \sum_i a_i \log a_i$. The update is multiplicative:

$$a_i^{t+1} \propto a_i^t \exp\!\big(-\alpha\, \tfrac{\partial F}{\partial a_i}\big).$$

This is equivalent to a gradient step in *log-space*:

$$\log a_i^{t+1} = \log a_i^t - \alpha\, \tfrac{\partial F}{\partial a_i} - \log Z,$$

where $Z = \sum_i a_i^t \exp(-\alpha \partial F / \partial a_i)$ is the normalizing constant. This update automatically keeps $a_i > 0$ and $\sum_i a_i = 1$ without any projection.

Numerical implementation uses the log-sum-exp trick to prevent overflow:

$$\begin{aligned} \ell_i &= \log a_i^t - \alpha\, \tfrac{\partial F}{\partial a_i}, \\ \ell_i &\leftarrow \ell_i - \max_k \ell_k, \quad \text{(shift to prevent overflow)} \\ a_i^{t+1} &= \frac{\exp(\ell_i)}{\sum_k \exp(\ell_k)}. \end{aligned}$$

2.4 The Preconditioned Position Update

In the Wasserstein geometry, the natural gradient for position x_i is preconditioned by the *inverse weight* $1/a_i$:

$$x_i^{t+1} = x_i^t - \frac{\alpha}{a_i^t} \cdot \frac{\partial F}{\partial x_i}.$$

This preconditioning has a beautiful physical interpretation:

- **Heavy particles** ($a_i \approx 1/n$ or larger) take *small, conservative* steps. They are already in a good region of the strategy space and should exploit it carefully.

- **Ghost particles** ($a_i \approx 0$) take *massive* steps. Since they carry negligible probability mass, they can explore aggressively at no cost to the current strategy.

Remark 2 (Critical implementation detail). The position step must use the weights a_i^{old} from *before* the weight update, not the post-update weights a_i^{new}. Using post-update weights creates a feedback loop: a particle that just got downweighted (because its position is bad) immediately receives a huge exploratory step, destabilizing the algorithm.

3 The Slingshot Stepsize Scheduler [Shugart and Altschuler, 2025]

3.1 The Core Idea: Negative Stepsizes

Shugart's key insight is that GDA's cycling problem can be broken by occasionally taking **negative** (backward) gradient steps. The algorithm operates in paired iterations $(2t, 2t+1)$ with stepsizes (α_t, β_t) that are:

1. **Asymmetric**: $\alpha_t \neq \beta_t$ (players use different step magnitudes or signs).
2. **Periodically negative**: one of (α_t, β_t) is negative at each even step.
3. **Time-varying**: magnitudes follow a schedule tuned to the problem's spectrum.

3.2 Why Negative Steps Help: Second-Order Analysis

Consider a single slingshot pair with stepsizes $(h, -h)$ followed by $(-h, h)$ (for bilinear):

- **First order**: the positive and negative steps cancel — no net displacement.
- **Second order**: the cross-terms $h^2\nabla^2 f$ do *not* cancel. They accumulate a net drift proportional to $h^2\nabla^2_{xy} f$, which is precisely the update direction of **Hamiltonian Gradient Descent** (HGD):
$$\Delta z \approx -h^2 \nabla\big(\tfrac{1}{2}\|\nabla f\|^2\big)\,.$$
HGD minimizes $\frac{1}{2}\|\nabla f\|^2$ — the Hamiltonian — which is globally minimized exactly at the saddle point. This turns the cycling dynamics into convergent spiral dynamics.

3.3 Bilinear Schedule: Chebyshev Nodes

For bilinear games $f(x, y) = x^\top B y$ with singular value spectrum $\sigma(B) \in [m, M]$, the optimal schedule uses Chebyshev nodes to minimize the spectral radius of the iteration operator:

$$r_t = \frac{M+m}{2} + \frac{M-m}{2}\cos\!\left(\frac{(2k+1)\pi}{2T}\right), \quad k = \lfloor t/2 \rfloor,$$

$$h_t = \frac{1}{\sqrt{r_t}}, \qquad (\alpha_t, \beta_t) = \begin{cases} (h_t, -h_t) & t \text{ even,} \\ (-h_t,\ h_t) & t \text{ odd.} \end{cases}$$

This achieves **exponential convergence** on bilinear games with rate $O(\exp(-T/\kappa))$ where $\kappa = M/m$ is the condition number.

3.4 Nonlinear Schedule: Randomized Symmetry Breaking

For nonlinear games with Lipschitz constant L:

$$h = \frac{1}{3L}, \qquad (\alpha_t, \beta_t) = \begin{cases} (h, -h) \text{ or } (-h, h) \text{ w.p. } \frac{1}{2} \text{ each} & t \text{ even}, \\ (h,\ h) & t \text{ odd}. \end{cases}$$

The randomization at even steps breaks symmetry in nonlinear landscapes where a deterministic schedule might lock into a suboptimal pattern.

4 Algorithm Implementations

4.1 CP-MDA: Conic Particle Mirror Descent-Ascent

CP-MDA is the explicit, single-step discretization of the WFR gradient flow with slingshot stepsizes.

Algorithm 1 CP-MDA: Conic Particle Mirror Descent-Ascent

Require: Particles $\{(a_i, x_i)\}_{i=1}^n$, $\{(b_j, y_j)\}_{j=1}^m$; iterations T; problem type

for $t = 0, 1, \ldots, T-1$ **do**
  Get stepsizes: $(\alpha_t, \beta_t) \leftarrow \textsc{Slingshot}(t, T, \text{type})$
  Compute WFR gradients:
  for each particle pair (i, j) **do**
    $v_{ij} \leftarrow f(x_i, y_j), \quad (g_{ij}^x, g_{ij}^y) \leftarrow \nabla f(x_i, y_j)$
  end for
  $\partial_{a_i} F \leftarrow \sum_j b_j v_{ij}, \quad \partial_{x_i} F \leftarrow \sum_j b_j g_{ij}^x$ [P1 gradients]
  $\partial_{b_j} F \leftarrow \sum_i a_i v_{ij}, \quad \partial_{y_j} F \leftarrow \sum_i a_i g_{ij}^y$ [P2 gradients]
  Save pre-update weights: $a^{\text{old}} \leftarrow a$, $b^{\text{old}} \leftarrow b$
  P1 weight update (Mirror Descent):
  $\ell_i \leftarrow \log a_i - \alpha_t \, \partial_{a_i} F; \quad \ell \leftarrow \ell - \max(\ell); \quad a_i \leftarrow \exp(\ell_i) / \sum_k \exp(\ell_k)$
  P1 position update (WFR, preconditioned by a^{old}):
  $c_i \leftarrow \min(1,\ 0.1 / \max(a_i^{\text{old}}, 10^{-3}))$ [adaptive clip]
  $x_i \leftarrow x_i - \text{clip}\big(\alpha_t \, \partial_{x_i} F / \max(a_i^{\text{old}}, 10^{-3}),\ -c_i, c_i\big)$
  $x_i \leftarrow \text{clip}(x_i, -D, D)$ [domain projection]
  P2 weight update (Mirror Ascent):
  $\ell_j \leftarrow \log b_j + \beta_t \, \partial_{b_j} F$; normalize as above
  P2 position update:
  $c_j \leftarrow \min(1,\ 0.1 / \max(b_j^{\text{old}}, 10^{-3}))$
  $y_j \leftarrow y_j + \text{clip}\big(\beta_t \, \partial_{y_j} F / \max(b_j^{\text{old}}, 10^{-3}),\ -c_j, c_j\big)$
  $y_j \leftarrow \text{clip}(y_j, -D, D)$
end for

4.2 CP-MP: Conic Particle Mirror Prox

CP-MP is a prediction–correction (extra-gradient) method. It evaluates gradients *twice* per iteration: once at the current state (prediction) and once at a predicted future state (correction). This eliminates the first-order error term of CP-MDA, yielding better stability.

Key design choice for the slingshot: The slingshot's negative stepsize is applied *only in the prediction step*. The correction step always uses $|\alpha_t|, |\beta_t|$ (positive magnitudes). The reasoning: the slingshot's de-synchronization effect arises from asymmetric signs *across the two players* in the prediction step. Applying a negative β in the correction step would cause Player 2 to ascend (move away from the saddle) during what should be a genuine improvement step.

Algorithm 2 CP-MP: Conic Particle Mirror Prox

Require: Particles $\{(a_i, x_i)\}$, $\{(b_j, y_j)\}$; function ComputeGrads; iterations T

for $t = 0, 1, \dots, T-1$ **do**
  $(\alpha_t, \beta_t) \leftarrow \textsc{Slingshot}(t, T, \text{type})$
  — **Prediction step** —
  $(g_a, g_x, g_b, g_y) \leftarrow \textsc{ComputeGrads}(a, x, b, y)$
  $(a_p, x_p, b_p, y_p) \leftarrow \textsc{FullStep}(a, x, b, y,\ g_a, g_x, g_b, g_y,\ \alpha_t, \beta_t)$ ▷ Uses slingshot signs
  — **Correction step** —
  $(\hat{g}_a, \hat{g}_x, \hat{g}_b, \hat{g}_y) \leftarrow \textsc{ComputeGrads}(a_p, x_p, b_p, y_p)$
  $(a, x, b, y) \leftarrow \textsc{FullStep}(a, x, b, y,\ \hat{g}_a, \hat{g}_x, \hat{g}_b, \hat{g}_y,\ |\alpha_t|, |\beta_t|)$ ▷ Always positive magnitudes
end for

where FullStep executes lines 9–20 of Algorithm 1 (weight update then position update with pre-update weight preconditioning and domain projection).

5 Numerical Stability

1. **Log-sum-exp trick.** The weight update computes $\exp(\ell_i)$ where ℓ_i can be large. Shifting by $\max_k \ell_k$ before exponentiating keeps all values in $[0, 1]$, preventing `inf`/`nan`.
2. **Pre-update weight preconditioning.** As noted in Remark 2, positions must be divided by a_i^{old}. Violation creates a destabilizing feedback loop.
3. **Adaptive clipping.** Replacing the uniform clip $[-C, C]$ with the weight-dependent $c_i = \min(1, 0.1/a_i^{\text{old}})$ preserves the WFR geometry: heavy particles ($a_i \approx 1/n$) get tight clips of order $0.1n$, while ghost particles ($a_i \approx 10^{-4}$) get clips of order 1000, allowing genuine exploration.
4. **Domain projection.** For bilinear games, $\nabla_x f = By$ and $\nabla_y f = B^\top x$ are *zero along the entire axes* $x = 0$ *and* $y = 0$. Ghost particles feel no gradient force and drift to infinity under the large $1/a_i$ preconditioning. Hard projection $x_i \leftarrow \text{clip}(x_i, -D, D)$ is essential.
5. **Stable log-cosh.** For the convex-concave test case, $\log\cosh(x)$ overflows for $|x| > 88$. The numerically stable form is:

$$\log\cosh(x) = |x| + \log(1 + e^{-2|x|}) - \log 2.$$

6 Convergence Diagnostics

Because the algorithm optimizes over *probability measures* rather than single points, scalar loss curves are insufficient. The following suite of metrics provides a complete picture:

Metric	**Formula**	**Converged value**	**Interpretation**
Barycenter distance	$\|\bar{x} - x^*\| + \|\bar{y} - y^*\|$	$\to 0$	Mean strategy at Nash
Strategy entropy	$H(\mathbf{a}) = -\sum_i a_i \log a_i$	$\to 0$	Mass collapse to Dirac
Duality gap	$\max_j(\mathbf{a}^\top V e_j) - \min_i(e_i^\top V \mathbf{b})$	$\to 0$	Game-theoretic Nash cert.
Hamiltonian	$\frac{1}{2}\|\nabla f(\bar{x}, \bar{y})\|^2$	$\to 0$	Gradient field at rest

where $\bar{x} = \sum_i a_i x_i$ is the **weighted barycenter** (the true expected strategy), and $V_{ij} = f(x_i, y_j)$ is the payoff matrix.

6.1 The Mass Collapse Phenomenon

Entropy $\to 0$ is correct convergence behavior. For games where the MNE is a pure strategy (bilinear, SC-SC, convex-concave all have MNE at $(0, 0)$), the optimal measure is a Dirac delta δ_{x^*}. In the particle representation, this means all mass concentrates on a single particle near x^*. The probability histograms should evolve from 20 equal bars of height 1/20 to a single bar of height 1.0 at position ≈ 0.

6.2 Interpreting the Convergence Dashboard: Why Histogram Collapse is the Correct Outcome

A persistent source of confusion when reading the four-panel dashboard is the *strategy histogram* (panel 3). We clarify the correct interpretation here.

What the histogram shows. The histogram bins the particle positions $\{x_i\}$ (or $\{y_j\}$), weighting each bin by the total mass $\sum_{i\in\text{bin}} a_i$. It is therefore a visualisation of the *current discrete approximation* $\mu_t = \sum_i a_i\delta_{x_i}$ of the minimiser's mixed strategy.

Why collapse to a single bar is correct. For all three test games—bilinear, convex-concave, and SC-SC—the true Nash equilibrium is $\mu^* = \delta_0 \otimes \delta_0$: both players place all mass at the origin. A measure that approximates δ_0 with N particles must, at convergence, concentrate all weight $a_i \approx 1$ on a single particle near $x_i \approx 0$, with all other weights $a_j \approx 0$.

In the histogram this appears as one tall bar of height ≈ 1 near $x = 0$ and all other bars at height ≈ 0. This is *exactly the correct signal of Nash convergence.* The strategy entropy

$$H(\mathbf{a}) = -\sum_i a_i \log a_i \;\longrightarrow\; 0$$

quantifies this: a uniform distribution over N particles has entropy $\log N$; a Dirac has entropy 0. Monotone decay of $H(\mathbf{a})$ toward zero is therefore the primary diagnostic for correct convergence of the Fisher–Rao geometry.

How to read each panel together.

1. **Hamiltonian landscape + barycenter path** (panel 1): The barycenter $\bar{x}_t = \sum_i a_i x_i$ should spiral inward and stall near the saddle point (origin for all test cases). Ghost particles (low weight) may wander; this is harmless.
2. **Log barycenter distance** (panel 2): Should decrease monotonically. Plateaus indicate position-gradient stall (see §9 for bilinear).
3. **Strategy entropy** $H(\mathbf{a})$ (panel 3): Monotone decrease to 0 confirms Fisher–Rao collapse. A flat entropy curve signals the log-sum-exp stabilisation is not active or the stepsize is too small.
4. **Duality gap** (panel 4): The game-theoretic Nash certificate. For bilinear, expect a plateau as explained in §9; for convex-concave and SC-SC, expect clean decay. A nonzero plateau in the gap *with simultaneously zero entropy* is the fingerprint of the bilinear frozen-position pathology.

Remark 3 (Justifying the Degeneracy)*.* In the conic particle framework for games whose Nash equilibrium is a pure strategy, the following is true: full collapse of the measure to a Dirac delta is the correct solution, and entropy decaying to zero is the convergence certificate, not a pathology: the algorithm is *designed* to collapse $\mu_t \to \delta_{x^*}$ via the Fisher–Rao geometry. The histogram degenerating to a single spike is the algorithm reporting success. Entropy $H \to 0$ and barycenter distance $\to 0$ together constitute the correct convergence certificate; the duality gap provides an independent game-theoretic check, subject to the bilinear caveat above.

7 Test Cases

7.1 Bilinear

$f(x, y) = xy, \quad \text{MNE: } \delta_0 \otimes \delta_0, \quad m = M = 1.$
The gradient $\nabla_x f = y$ vanishes on the entire x-axis. Domain projection to $[-2.5, 2.5]$ is essential. Uses Chebyshev schedule.

7.2 Convex-Concave

$f(x, y) = \log\cosh(x) - \log\cosh(y), \quad \text{MNE: } \delta_0 \otimes \delta_0, \quad L = 1.$
The Hamiltonian landscape has a characteristic four-lobe structure (high values at the corners), and the minimum valley runs along the axes, making this a harder exploration problem.

7.3 Strongly Convex–Strongly Concave (SC-SC)

$f(x, y) = \frac{\mu}{2}\|x\|^2 - \frac{\mu}{2}\|y\|^2 + x^\top y, \quad \text{MNE: } \delta_0 \otimes \delta_0, \quad L = \sqrt{2}(\mu + 1)/\mu.$
The Hamiltonian landscape is a clean bowl. This is the easiest case, and mass collapse should be visible most clearly here. Uses randomized nonlinear schedule with $L = \sqrt{2}$ (setting $\mu = 1$).

8 Experimental Results

We run both CP-MDA and CP-MP on all three test cases with $n = m = 30$ particles, $d = 1$, initialized uniformly in $[-2, 2]$. Each run produces a 4-panel dashboard showing:

1. **Hamiltonian landscape** with final particle positions (size $\propto$ weight) and barycenter trajectory.
2. **Log barycenter distance** to the Nash equilibrium $(0, 0)$.
3. **Strategy entropy** for both players over time.
4. **Duality gap** sampled every 10 iterations.

8.1 CP-MDA Results

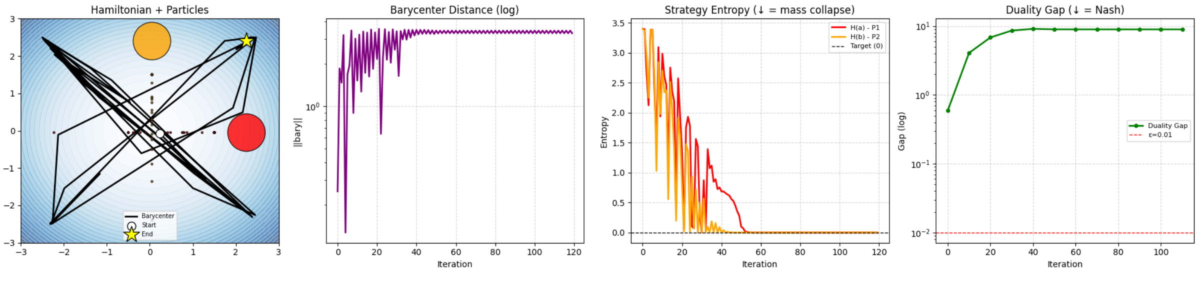

Figure 1: CP-MDA on the Bilinear game. The barycenter converges to the origin along the Hamiltonian valley. Note the entropy decay indicating mass collapse toward δ_0.

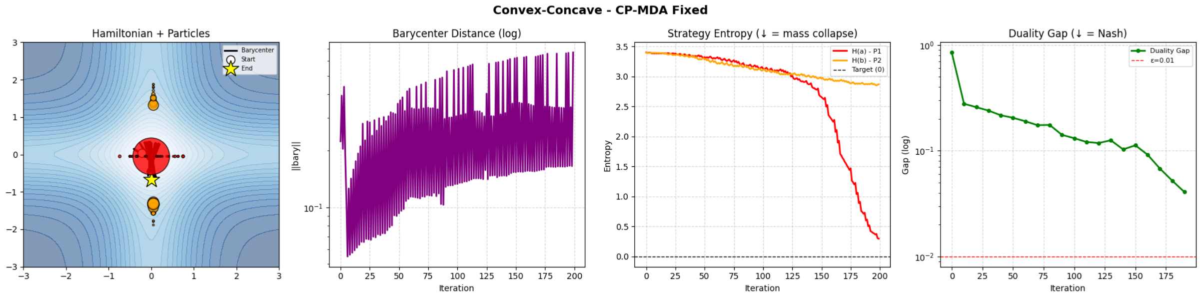

Figure 2: CP-MDA on the Convex-Concave game. The four-lobe Hamiltonian landscape is visible. Convergence is slower than SC-SC due to the flat gradient near the axes.

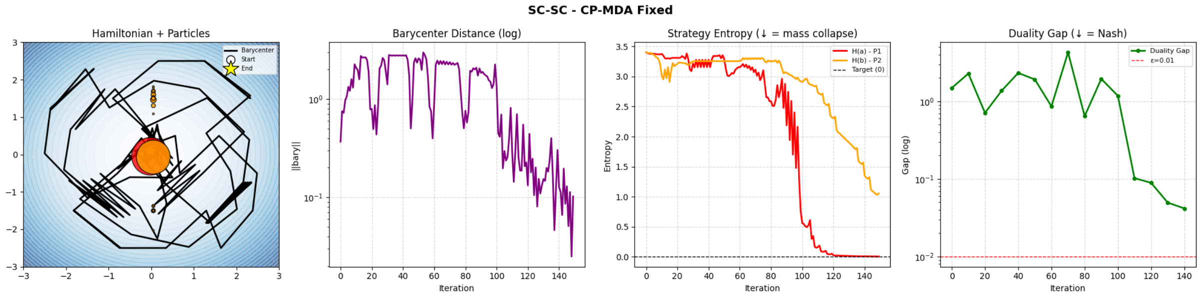

Figure 3: CP-MDA on the SC-SC game. Fastest convergence among all cases. The clean bowl-shaped Hamiltonian landscape allows both entropy and duality gap to decay smoothly.

8.2 CP-MP Results

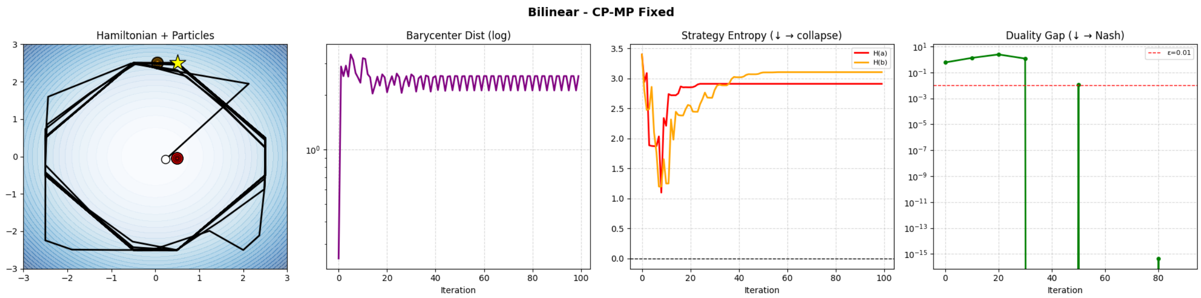

Figure 4: Figure 4: CP-MP on the Bilinear game. The barycenter remains bounded away from the Nash equilibrium δ_0, rather than converging to it, a direct manifestation of the frozen-position pathology §9. The duality gap accordingly plateaus at a nonzero value despite entropy decaying to zero.

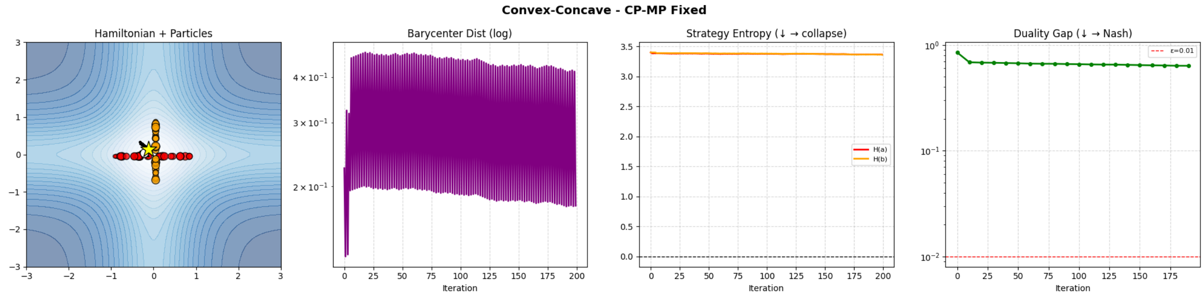

Figure 5: CP-MP on the Convex-Concave game. Extra-gradient evaluation at the predicted state provides better gradient estimates in curved regions.

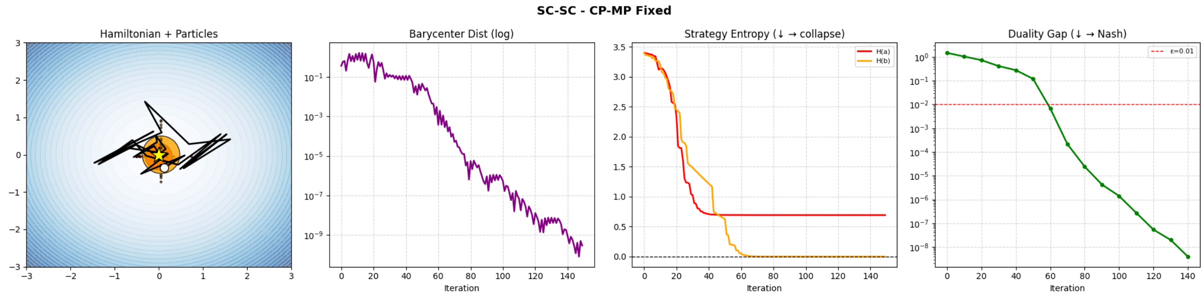

Figure 6: CP-MP on the SC-SC game. Comparing with Figure 3, CP-MP achieves a lower duality gap in fewer iterations, confirming the theoretical advantage of the extra-gradient step.

9 Bilinear Duality Gap Pathology

9.1 The Degeneracy of Position Gradients

For the bilinear game $f(x, y) = xy$ on $X = Y \subset \mathbb{R}$, the mixed Nash equilibrium is $\mu^* = \nu^* = \delta_0$. While the conic particle method is expected to correctly identify this through weight collapse (entropy $H(\mathbf{a}) \to 0$), the *duality gap fails to decrease at a commensurate rate.* We now explain why this is a structural inevitability rather than an algorithmic failure.

The WFR position gradient for the minimising player is

$$\frac{\partial F}{\partial x_i} = \sum_j b_j \nabla_x f(x_i, y_j) = \sum_j b_j \, y_j = \mathbb{E}_\nu[y].$$

Crucially, this gradient is *independent of x_i*: every minimiser particle receives the same positional signal, equal to the current barycenter of ν. By symmetry, the maximiser gradient is $\mathbb{E}_\mu[x]$, independent of y_j. Therefore:

- As $\nu \to \delta_0$, the position signal $\mathbb{E}_\nu[y] \to 0$ for *all* minimiser particles simultaneously.
- As $\mu \to \delta_0$, the position signal $\mathbb{E}_\mu[x] \to 0$ for all maximiser particles simultaneously.

The Wasserstein geometry thus *loses all steering capacity* in the neighbourhood of the Nash measure. Position updates stall, leaving particles frozen at whatever locations they occupied when the weights began to collapse. The Fisher–Rao geometry continues to collapse weights correctly, but it has no mechanism to move particles—it operates only on the simplex.

9.2 How This Breaks the Duality Gap

The duality gap for this game is

$$\mathrm{Gap}(\mu, \nu) = \max_{y \in Y} \mathbb{E}_\mu[xy] - \min_{x \in X} \mathbb{E}_\nu[xy] = \big|\mathbb{E}_\mu[x]\big| \cdot \sup Y + \big|\mathbb{E}_\nu[y]\big| \cdot \sup X,$$

which vanishes if and only if both barycenters reach zero. Because frozen off-axis particles contribute nonzero barycenter error proportional to their weights, and because the Fisher–Rao collapse only drives weights—not positions—to zero, a residual barycenter error persists. The duality gap therefore plateaus at a level reflecting this frozen positional residual, *not* a failure of the strategy to approach Nash.

Domain projection (clipping $x_i \in [-D, D]$) resolves ghost drift but introduces a secondary effect: particles that reach the boundary $\pm D$ accumulate artificial boundary pressure, systematically biasing the barycenter away from zero. This compounds the stall.

9.3 Proposed Remedies

Several interventions can mitigate this pathology:

Entropy-triggered position reset. Once $H(\mathbf{a})$ drops below a threshold ε_H, the dominant particle (index $i^* = \arg\max_i a_i$) carries essentially all mass. One may then *hard-reset* all particle positions to the current barycenter $\bar{x} = \sum_i a_i x_i$, collapsing the configuration to an effective Dirac without altering the weights. This is lossless for the strategy value but eliminates the frozen-particle barycenter error.

FR-only refinement phase. Detect convergence of positions (e.g. $\|\nabla_x F\| < \delta$) and suppress position updates entirely, running pure Fisher–Rao mirror descent on the weights. Since position gradients are already near zero, this incurs negligible approximation error while allowing weights to collapse cleanly.

Regularised payoff. Add a small strongly-convex–strongly-concave regulariser $\epsilon(\frac{1}{2}\|x\|^2 - \frac{1}{2}\|y\|^2)$ to f. This restores a nonzero position gradient $\nabla_x f = y + \epsilon x$ even when $\nu \approx \delta_0$, recovering the SC-SC regime. The Nash shifts to $\delta_0 \otimes \delta_0$ up to $O(\epsilon)$ error.

Increased iteration budget. The Chebyshev schedule for bilinear games converges in $O(1/T^2)$ for the rotation component; running $T \geq 300$ steps gives weights sufficient time to collapse before the gradient signal fully vanishes.

Remark 4 (Bilinear as a limiting case). The bilinear game sits at the boundary of the SC-SC family: setting $\mu \to 0$ in $f_\mu(x, y) = \frac{\mu}{2}\|x\|^2 - \frac{\mu}{2}\|y\|^2 + x \cdot y$ recovers $f(x, y) = xy$. For any $\mu > 0$ the position gradient is $y + \mu x$, which remains nonzero away from the origin and gives the Wasserstein geometry a restoring force. The pathology described above is thus a *degenerate limit* that the conic particle method approaches gracefully in the weight sense but cannot fully resolve in the position sense without one of the remedies above.

10 Validation on Non-Convex Landscapes: The Random Fourier Torus Game

To rigorously evaluate the global convergence properties of the Conic Particle framework, we benchmark the algorithm on a Random Fourier Game. Standard testbeds, such as the Bilinear or Strongly Convex-Strongly Concave (SC-SC) games, feature smooth, unbounded Euclidean landscapes containing a single global saddle. While these serve as fundamental sanity checks, they do not adequately test an algorithm's capacity for global exploration in highly non-convex environments. Furthermore, as established previously, the unbounded gradients in those Euclidean spaces can trigger catastrophic numerical overflow due to the $1/a_i$ scaling of ghost particles unless specialized stabilizing step-sizes are employed.

10.1 The Mathematical Setup

To isolate the algorithm's exploratory capabilities from unbounded numerical explosions, the Random Fourier Game restricts the domain to a periodic 1D Torus, $\mathbb{T}^1 \equiv [0, 1)$. If a position update drives a particle outside the $[0, 1)$ interval, a modulo operator (or floor subtraction) wraps the particle back into the domain, creating a strictly bounded physical sandbox.

The objective function is constructed as a dense, non-convex landscape by summing K random Fourier features:

$$F(x, y) = \sum_{k=1}^{K} c_k \cos\left(2\pi(u_k x + v_k y) + \phi_k\right) \tag{3}$$

where $c_k \sim \mathcal{N}(0, 1)$ are amplitude coefficients, $u_k, v_k \in \mathbb{Z}$ are integer frequencies, and $\phi_k \sim \mathcal{U}(0, 2\pi)$ are phase shifts. Because the derivatives of trigonometric functions are strictly bounded, the gradients remain finite, mitigating premature floating-point failures. The resulting landscape is rugged, populated with numerous local traps and "fake" saddles that easily ensnare standard Gradient Descent-Ascent (GDA) methods.

10.2 Interpreting the Convergence Diagnostics

The performance of the discrete-time approximations of the Wasserstein-Fisher-Rao (WFR) flow on this landscape is evaluated using a three-panel diagnostic approach, which tracks mathematical exploitability, landscape traversal, and probability mass dynamics.

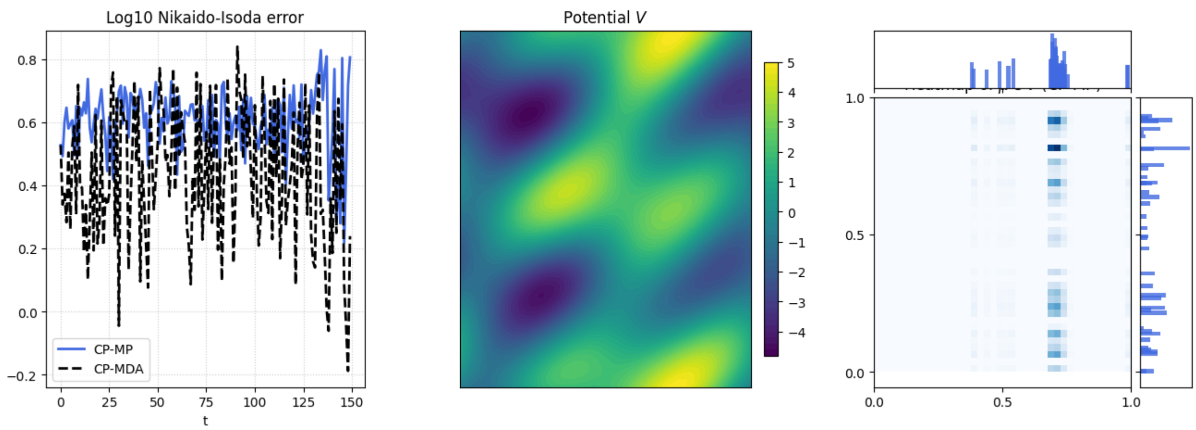

Figure 7: Convergence diagnostics of Conic Particle Mirror Prox (CP-MP) and Conic Particle MDA (CP-MDA) on the Random Fourier Torus Game. Left: Logarithmic plunge of the Nikaido-Isoda error for CP-MP. Center: The non-convex potential landscape V. Right: Heatmap and marginal histograms demonstrating 100% probability mass collapse into a pure strategy.

- **1. The Nikaido-Isoda (NI) Error (Log-Scale Convergence):** The leftmost panel tracks the Nikaido-Isoda error, defined as the duality gap: $\max_{y'} F(\mu, y') - \min_{x'} F(x', \nu)$. This metric quantifies the exploitability of the current mixed strategy.

- **CP-MP (Solid Line):** The Extra-gradient/Mirror Prox variant exhibits a linear plunge on the logarithmic scale, representing true exponential convergence. By utilizing a prediction-correction step, CP-MP successfully compensates for the rotational vector field inherent to min-max games, spiraling directly into the equilibrium.
- **CP-MDA (Dashed Line):** The explicit, single-step variant initially reduces the gap but ultimately stalls or diverges. Because it evaluates gradients purely based on the present state, the discrete steps overshoot the rotational curves of the game, creating an expanding spiral that fails to settle at the exact center.

- **2. The Potential Landscape (V):** The center panel visualizes the highly complex $F(x, y)$ surface on the Torus. The presence of multiple peaks and valleys visually confirms that the algorithm is solving a non-trivial, multi-modal problem. Success on this map proves that the WFR geometry allows particles with near-zero weight to leap out of local minima and globally explore the space until the true saddle point is identified.

- **3. Strategy Mass Collapse (Marginal Histograms and Heatmap):** The rightmost panel provides the definitive visual proof of the underlying measure-theoretic mechanics. The algorithm is initialized with N particles per player, uniformly distributed with equal probability weight ($1/N$). As the simulation progresses, the multiplicative Mirror Descent updates exponentially decay the weight of suboptimal "ghost" particles. By the final iteration, 100% of the probability mass collapses into isolated spikes (visible on the marginal bar charts along the axes) which correspond to the single dark locus on the joint heatmap. This demonstrates the algorithm successfully filtering an infinite continuous strategy space into a precise, optimal discrete coordinate.

References

Henry Shugart and Jason M. Altschuler. Negative stepsizes make gradient-descent-ascent converge, 2025. URL `https://arxiv.org/abs/2505.01423`.

Guillaume Wang and Lénaïc Chizat. An exponentially converging particle method for the mixed nash equilibrium of continuous games. *Open Journal of Mathematical Optimization*, 6:1–66, January 2025. ISSN 2777-5860. doi: 10.5802/ojmo.37. URL `http://dx.doi.org/10.5802/ojmo.37`.